

Face recognition login

Brillertek has developed a Face Recognition software for Windows platforms that allows users to login using just biometric recognition

Trace your family

Automated face recognition technology can potentially be used to research your genealogy by crawling the web for similar faces

and is one of the biggest challenges in the field today.

In the Face Recognition Grand Challenge held in 2006 at Arlington, Virginia, the new algorithms proved to be ten times more precise than those of 2002. The advances even permitted the system to accurately identify and differentiate between identical twins.

Skin texture

Skin biometrics is another supportive technology being developed by companies such as Identix, which merged with identity solutions provider, Viisage in 2006 – using the uniqueness of skin texture for more precise results. The software takes a picture of a patch of skin – a *skinprint*, which is broken into smaller portions and converted into a mathematical space by an algorithm, picking the lines, pores and patterns that constitute the skin. Systems such as FaceIt have been developed to incorporate Eigenvectors, local feature analysis and surface texture analysis to optimise results. However, long hair, dim lights or sunglasses could hinder the system's performance.

The commitment of resources by governments and venture capitalists and the labour of scientists and research students indicate face recognition is developing in the most unanticipated directions and face biometrics has already become an integral part of our daily life.

Tomorrow is today

It is the distinctive feature of our times that we shall live to see many science fiction predictions come true in our own lifetimes. A lot of people were impressed by the sci-fi gadgetry in Steven Spielberg's *Minority Report*.

If you were one of them, you'd probably be even more impressed, and pleasantly surprised to know that something quite similar is already happening around you. Here are a few examples to shock and awe you:

Infrared facials

A new innovative project in the UK has led to a breakthrough in eSecurity services by using infrared light. St. Neots Community College, in Cambridgeshire has given its support to the project and allowed the system to be installed in order to record

attendance. The students only just need to walk over to a scanner, and enter a PIN for their identity to be verified. The scanner is basically a system that takes an image using infrared light, invisible to the human eye, reinforcing the belief that face recognition is the most non-intrusive and time-saving biometric technology today. Poor ambient lighting does not hamper the system because the infrared flash lights up the face sufficiently, without the subject perceiving it. The system measures features such as distance between the pupils of the eyes, and, in less than 2 seconds, makes a match with the templates in the existing database.

Korean forays

San Francisco-based 3VR Security Inc. recently released the results of its facial recognition technology tests, conducted in collaboration with Korea's SK Networks. The report claims 85 to 92 per cent accuracy with very few false positives, in the most uncontrolled and highly populated environments viz. the subway stations in Seoul. 3VR's special software is geared towards overcoming the challenges of video-based biometrics, specifically, the task of recognising and tracking faces in crowds of hundreds in real time, in spite

of poor lighting and imperfect camera angles. If 3VR's re-modelled still-image face recognition algorithms find their way into CCTV cameras all over the world, who knows, in a few years, you might just be picked off the street for that age-old parking ticket you didn't bother to pay.

Faucet facets

iHome, a company based in China has come out with the ultimate luxury device – a super-intelligent tap! The *Smartfaucet* activates itself once it detects motion and uses face recognition technology to identify who exactly has stepped into the tub. The device then sets the temperature of the bath according to your prior preferences. It even has an LCD touch screen that can connect you to the world wide web.

Face-savvy facades

Yahoo! Japan plans to install billboards that scan passers-by, identify their gender and age group, using facial recognition technology, and in true Minority Report style flash advertisements based on this assessment. Yahoo! has teamed up with Comel – a Tokyo based company – for the hardware, and NEC Soft (Japan) for the facial analysis technology. The boards will also display weather updates and news content apart from advertisements. Yahoo plans to set up about 500 such billboards in malls and railway stations of Fukoka, to begin with and later expand to Tokyo



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Smarter licences

The transport authorities in Queensland, Australia have decided to replace almost three million drivers' licenses with 'smartcards' using facial recognition technology provided by Unisys Australia. The project is huge, involving several other partner companies such as Leigh Mardon Australia that will design the customer interface devices, and Daon, which will provide the biometric enrolment technology and middleware

software. Starting in April 2009, and over the period of the next five years, Unisys will help prevent identity theft by embedding a biometric facial image into the smartcard chip. Typically, if someone's photo does not match the one on the card or matches the picture on another card – an alarm

and Osaka. And you don't need to wake up when September ends for this one – it's already happening.

Lip Reads

Ever since the London bombings and the hours and hours of CCTV footage that had to be examined, law enforcers in the U.K. have decided to take the bull of terrorism by the horns. Earlier this year, groundbreaking research has begun, in an attempt to rectify the aberrations of suspect identification on CCTV records, by adding lip movement and speech pattern recognition to existing face recognition software. Face recognition experts, Omniperception and BAE Systems, a defence and security company, have tied up to try and hone the system to such a degree that a subject's face could be accurately recognised from distances as large as a hundred metres.

It looks like the entire world seems to be working to create that perfect face recognition solution. But what lies in store for face-based software in the near future? In order to remedy the inaccuracies of the systems we've developed so far, scientists all over the world have realised that the solution lies not just in 'face-recognition' but in other branches of science entirely, namely, neurophysiology, ornithology, entomology and the human visual system.



Minority Report had showcased high tech human identification technologies